Precision Agriculture (2025) 26:17
https://doi.org/10.1007/s11119-024-10213-1

Drivers and barriers to precision agriculture technology and digitalisation adoption: Meta-analysis of decision choice models

Zdeňka Žáková Kroupová[1] · Renata Aulová[1] · Lenka Rumánková[1] · Bartłomiej Bajan[1,2] · Lukáš Čechura[1] · Pavel Šimek[3] · Jan Jarolímek[3]

Accepted: 3 December 2024

Abstract
The article defines the key determinants of adopting precision agriculture technologies and digitalisation. The research objectives are fulfilled by the systematic review and meta-analysis of relevant studies, identified and selected in accordance with the PRISMA protocol in the Web of Science and Scopus databases. The findings emphasize the importance of socio-economic factors, such as education, age, and farm size. High technical literacy and adequate information about new technologies—including their expected profitability—are crucial for assessing the benefits of precision agriculture and digitalisation, on which a more considerable expansion of these technologies into the practice of agricultural entities depends. Large and capital-intensive enterprises are more likely to implement new technologies in production practices, especially if they are led by younger and more educated managers who are more open to modern technologies and are more willing to take risks.

Keywords Precision farming · Digitalisation · Adoption · Meta-analysis · PRISMA

Introduction

The 21st century brings several technological innovations that are radically changing the production process in agriculture by introducing technologies that improve the decision-making process of agricultural producers, as well as increasing the efficiency of mechanisa-

✉ Zdeňka Žáková Kroupová
kroupovaz@pef.czu.cz

[1] Department of Economics, Faculty of Economics and Management, Czech University of Life Sciences Prague, Kamýcká 129, 165 00 Prague, Czech Republic

[2] Department of Economics and Economic Policy in Agribusiness, Faculty of Economics, Poznań University of Life Sciences, Wojska Polskiego 28, 60-637 Poznań, Poland

[3] Department of Information Technologies, Department of Economics, Czech University of Life Sciences Prague, Kamýcká 129, 165 00 Praha 6, Czech Republic

Published online: 27 December 2024

tion and the extent of automation of agricultural production. Timely and accurate decisions as a result of the implementation of diagnostic and analytical tools (e.g. soil mapping, yield monitoring) and artificial intelligence enable the optimisation of inputs according to agronomic requirements, and together with precise application technologies (e.g., variable rate application) increase the efficiency and sustainability of agricultural production (Troiano et al., 2023). Robotics and autonomous technologies also respond to labour shortages in agriculture (Tey & Brindal, 2022) and, together with other new technologies, make agricultural work more attractive (Torero, 2021).

These technologies, summarised in the literature under the designations: precision agriculture, smart farming, digital farming, and agriculture 4.0 (Klerkx et al., 2019), have numerous social, economic, and environmental benefits (Blasch et al., 2022). However, their diffusion and adoption by agricultural producers is still insufficient and slow (Troiano et al., 2023). The causes of this situation can be found in insufficient informedness, lack of funds for investment in new technologies, and the short economic lifespan of investments (e.g., due to incompatibility of equipment) (Blasch et al., 2022). The negative aspects of this situation manifest themselves not only on an economic (e.g., loss of competitiveness) and environmental (e.g., higher carbon footprint from agricultural production) level but also on a socio-demographic level (e.g., the unwillingness of young people to work in agriculture, and the associated depopulation of the countryside). They call for a solution on a political level.

Implementing precision/digital agriculture technologies in production represents a critical economic decision for agricultural producers, as achieving potential economic and environmental benefits often requires a considerable financial investment (Mathanker, 2021). The decision-making process is influenced by factors such as expected cost savings, anticipated yield increases, and the resulting expected profitability and return on investment, as well as the producer’s risk tolerance, openness to new or environmentally friendly technologies, and the level of public support (D’Antoni et al., 2012). It is evident that economic factors do not solely drive the decision but are also influenced by social and political considerations that shape the agricultural producer’s decision to adopt precision/digital agriculture technologies (Khanal et al., 2019).

Different agricultural producers tend to adopt precision agriculture technologies and digitalisation[1] to different extents. Current research indicates that absorptive capacity (i.e., the ability of an enterprise to recognize, assimilate, and apply a new technology) is affected mainly by the size of the enterprise, the age and education of the manager, access to funds (income), and access to information. The probability of adopting precision agriculture technologies is generally estimated to be higher for large agricultural producers with higher incomes and easier access to capital compared to small farms (Barnes et al., 2019). Younger, more educated farmers tend to be more interested in new technologies and less risk-averse, making them earlier and more frequent adopters of precision agriculture technologies and digitalisation than older farmers. Education and dissemination of information play an unquestionable role in this process (Yarashynskaya & Prus, 2022).

[1] The term digitization includes technologies such as big data, the Internet of Things, augmented reality, robotics, sensors, 3D printing, artificial intelligence, machine learning and blockchain. In agriculture, digitization is often seen as part of the concepts of precision agriculture, digital agriculture, smart agriculture and agriculture 4.0 (Klerkx et al., 2019).

Understanding the drivers and barriers to adopting precision agriculture and digitalisation is crucial from microeconomic and political perspectives. This insight enables policymakers to focus their interventions on the most influential factors affecting the adoption of these technologies. Consequently, it facilitates the formulation of purposeful and effective support mechanisms to enhance the diffusion and implementation of these advanced agricultural technologies (Tey & Brindal, 2022) and achieve goals in economic, social, and environmental fields.

This study aims to (i) identify factors influencing the adoption of precision agriculture technologies and digitalisation, and (ii) evaluate the importance and significance of these factors in the adoption process. The findings can assist policymakers and technology developers in formulating more effective policies and strategies for the widespread implementation of these technologies. To achieve these objectives, a systematic review and the meta-analysis of relevant studies will be conducted.

The rest of the paper is organized as follows: The next sections present the current state of the art and the contribution of this study. The methodology section describes the materials and methods used for both qualitative and quantitative synthesis. The results section reports the findings of the qualitative synthesis first, followed by the results of the meta-analysis. The final section provides a summary of the key findings.

Current state of the art

Precision agriculture and digitalisation have been at the centre of attention for the scientific community for several decades. The importance and topicality of this subject are evidenced by over 16,900 records[2] in the Web of Science (WOS) database and 26,700 records[3] in the Scopus database, with almost 60% of these records from the last five years (i.e., 2019–2023). Non-review studies dominate this set of publications. In recent years, however, there has also been an increasing number of studies that synthesise existing research results (almost 70% of these studies were published in the last five years). The keywords adoption, implementation, use, determinants, factors, drivers, barriers, and motivation[4] narrow down the set of identified studies to 1,808 in the Web of Science and 3,252 in the Scopus database. This set can also be characterised by a predominant representation of non-review studies (78%) and studies published in the last five years (63%; this share increases to 73% if we include publications from the ongoing year 2024). Adding a query regarding systematic research or

[2] Searched by SQL query: TS = (“precision agriculture” OR “digital agriculture” OR “smart agriculture” OR “precision farming” OR “digital farming” OR “smart farming” OR “agriculture 4.0”) in the WOS database on 23 July 2024.

[3] Searched by SQL query: TITLE-ABS-KEY (“precision agriculture” OR “digital agriculture” OR “smart agriculture” OR “precision farming” OR “digital farming” OR “smart farming” OR “agriculture 4.0”) the Scopus database on 23 July 2024.

[4] Searched by SQL query: TS = (“precision agriculture” OR “digital agriculture” OR “smart agriculture” OR “precision farming” OR “digital farming” OR “smart farming” OR “agriculture 4.0”) AND TS = (adopt* OR implement* OR use*) AND TS = (“factors” OR “drivers” OR “determinants” OR “barriers” OR motivat*) in WOS and TITLE-ABS-KEY(“precision agriculture” OR “digital agriculture” OR “smart agriculture” OR “precision farming” OR “digital farming” OR “smart farming” OR “agriculture 4.0”) AND TITLE-ABS-KEY(adopt* OR implement* OR use*) AND TITLE-ABS-KEY(“factors” OR “drivers” OR “barriers” OR “determinants” OR motivat*) in the Scopus database on 23 July 2024.

meta-analysis[5] further reduces the number of studies to 54 in the Web of Science and 67 in the Scopus database, or more precisely, a total of 77 non-duplicated studies, whereby most of these studies (91%) have been published since 2019. The final number of 13 studies, as shown in Table 1, which realistically focus (according to the abstract and the complete text) on synthesising findings about determinants of adopting precision agriculture and digitalisation, highlights an existing research gap.

To date, over 100 determinants of adoption - both drivers and barriers - have been evaluated in systematic reviews and meta-analyses. These studies primarily find their theoretical foundation in Rogers's Theory of Diffusion of Innovations (Rogers, 2003), Davis's Technology Acceptance Model (Davis et al., 1989), Venkatesh's Unified Theory of Acceptance and Use of Technology (Venkatesh et al., 2003) and Grennhalgh's Model of Determinants of Diffusion (Grennhalgh et al., 2004), Dissemination and Implementation of Innovations (Sood et al., 2022; Degieter et al., 2023). The determinants can be categorized into technological, socioeconomic, informational, institutional, and political factors. Technological factors encompass the characteristics of the new technology, particularly its relative advantage, such as the benefit of the new technology compared to existing technologies in terms of efficiency, cost, and ecofriendliness. Other aspects include ease of use, compatibility, trialability (i.e., the opportunity for potential adopters to use an innovation before deciding to adopt) of the new technology, technological reliability and financial burden. Socioeconomic factors represent adopters' characteristics such as age, education, technical literacy, the size of the enterprise, and its economic situation. These factors affect the quality of human capital and the availability of financial capital. Therefore, some authors (Yarashynskaya & Prus, 2022; Sood et al., 2022) divide this broad category into social factors (education and age) and financial factors (income from the enterprise and availability of capital). Informational factors describe the dissemination of information (quality of communication channels, interconnectedness with other farmers, availability of expert advice) and technical support. Institutional factors mainly represent the availability of credit services and consulting. Political factors reflect public support for the diffusion and implementation of new technologies.

Degieter et al. (2023) and Bulut and Wu (2023) offer an alternative classification, dividing adoption determinants into internal and external. Internal factors are farm-specific and are represented by socio-demographic data (age, gender, education, experience), characteristics of agricultural enterprises (specialisation, size of enterprise, land ownership, indebtedness), perception of new technologies by adopters and their knowledge (expected economic and environmental benefits, enthusiasm for new technologies, technological literacy, expected difficulty of using the new technologies). External factors represent the sociopolitical environment in which the adopter of the new technology carries out their activity, and include characteristics of cooperation with nearby agricultural producers, innovative behaviour of nearby producers, dissemination of information and political interventions, and characteristics of the technology – its cost, compatibility, safety and availability of technical sup-

[5] Searched by SQL query: TS = ("precision agriculture" OR "digital agriculture" OR "smart agriculture" OR "precision farming" OR "digital farming" OR "smart farming" OR "agriculture 4.0") AND TS = (adopt* OR implement* OR use*) AND TS = ("factors" OR "drivers" OR "determinants" OR "barriers" OR motivat*) AND TS = ("systematic review" OR "meta-analysis") in WOS and TITLE-ABS-KEY ("precision agriculture" OR "digital agriculture" OR "smart agriculture" OR "precision farming" OR "digital farming" OR "smart farming" OR "agriculture 4.0") AND TITLE-ABS-KEY (adopt* OR implement* OR use*) AND TITLE-ABS-KEY ("factors" OR "drivers" OR "barriers" OR "determinants" OR motivat*) AND TITLE-ABS-KEY ("systematic review" OR "meta-analysis") in the Scopus database on 23 July 2024.

Table 1 Overview of systematic reviews and meta-analyses regarding precision agriculture adoption determinants

Author (year)	Journal	Number of primary studies	Source databases	Method (number of analysed determinants)	Techno-logical framework	Key adoption determinant
Bulut & Wu (2024)	Internet Research	51	AISeL, Emerald Insight, IEEE, ProQuest, Sage Journals, Science Direct, Taylor & Francis, Web of Science, Wiley Online	Systematic review (17)	Internet of Things	Cost, skills, standardization, connectivity, security
Cisternas et al. (2020)	Computers and Electronics in Agriculture	19	Scopus, Springer Link, Science Direct, Google Scholar, National Agricultural Library	Systematic review (N/A)	Information technology	Required knowledge
Degieter et al. (2023)	Agronomy Journal	23	Web of Science and Scopus	Systematic review (66)	Robotics and unmanned vehicles	Age, education, gender, farm size, perceived usefulness, expected economic and environmental benefits, perceived ease of use, attitude of confidence, financial condition, compatibility, and price of technology.
John et al. (2023)	Outlook on Agriculture	29	Web of Science and Scopus	Systematic review (N/A)	Precision agriculture	Awareness and perceptions, knowledge dissemination, cultural norms, initial investment costs, profitability, access to capital, technology literacy, availability of infrastructure, compatibility, and resource scarcity.

Table 1 (continued)

Author (year)	Journal	Number of primary studies	Source databases	Method (number of analysed determinants)	Techno-logical framework	Key adoption determinant
Lee et al. (2021)	Sustainability	31	Scopus	Systematic review (N/A)	Precision agriculture	Relative advantage, profitability, yield and quality enhancement, compatibility
Liu et al. (2021)	Journal of Cleaner Production	200	Web of Science	Systematic review (N/A)	Information-communication technology (ICT) and blockchain	Undefined
Osrof et al. (2023)	Technology in Society	182	Scopus, Google Scholar	Systematic review (112)	Smart farming	Farm size, age, education, financial costs, perceived benefits, income from farming, farm location, current technologies on farms, information sources, gender, technology literacy
Pathak et al. (2019)	Precision Agriculture	34	Scopus	Systematic review (64)	Precision agriculture	Relative advantage, adopter's characteristics
da Silveira et al. (2021)	Computers and Electronics in Agriculture	50	Science Direct, Web of Science, and Scopus	Systematic review (25)	Agriculture 4.0	Complexity, compatibility, reliability, technical literacy
Sood et al. (2022)	Journal of Decision Systems	65	Scopus, Web of Science, and ACM	Systematic review (16)	Artificial intelligence	Performance expectancy, effort expectancy, compatibility, social influence, information availability, facilitating conditions, financial condition, farm size, age, education and experience, risk aversion, formal and informal links, security and privacy

Table 1 (continued)

Author (year)	Journal	Number of primary studies	Source databases	Method (number of analysed determinants)	Techno-logical framework	Key adoption determinant
Tey and Brindal (2022)	Precision Agriculture	23 (qualitative synthesis), 18 (quantitative synthesis)	Google Scholar, Scopus, Web of Science	Meta-analysis of Hedge's g and weighted pooled effect (13)	Precision agriculture	Perceived profitability, use of computers, consulting
Yarashynskaya and Prus (2022)	Agronomy	21	Google Scholar, Science Direct, Scopus, Web of Science	Systematic review (9)	Precision agriculture	Age, education, farm size, farm income, credit availability, ICT equipment, internet availability, knowledge of technologies
Yatribi (2020)	Economics	41	Scopus, Science Direct	Systematic review (21)	Precision agriculture	Perception utility, farm size, education, ease of use

Source: own processing

Note: N/A indicates that the number of analysed factors cannot be accurately defined

port. Finally, Osrof et al. (2023) integrate these approaches by classifying the drivers and barriers of smart farming adoption into four categories: individual, organizational, technological, and external factors. Individual factors include age, education, gender, technology literacy. Oragnisational factors encompass farm size, farm location, income from farming, skilled labour. Technological are perceived benefits, compatibility, performance expectancy, perceived ease to use, perceived complexity. External factors involve information sources, social influence, government support, technical support.

One of the earliest studies synthesizing findings on these determinants to identify key factors in the adoption of precision agriculture technologies was conducted by Pathak et al. (2019). This study is conceptually based on the Model of Determinants of Diffusion, Dissemination, and Implementation of Innovations (Grennhalgh et al., 2004) and includes a large number of adoption determinants, such as relative advantage, compatibility, ease of use, trialability, the tangibility of result, existence of technical support, risk and nature of required knowledge (tacit/codified), social networks, sociopolitical climate, adopter's age and education, size of the enterprise, etc. Moreover, it defines as key factors for the adoption of precision agriculture technologies, relative advantage and adopter's characteristics (age, education, risk aversion level, size of enterprise, financial situation), which the authors summarise under the term "motivation". Similar key factors were also identified by Sood et al. (2022) and Degieter et al. (2023), highlighting the Unified Theory of Acceptance and Use of Technology (Venkatesh et al., 2003).

However, these findings are only partially supported by Tey and Brindal (2022). Their study is the sole meta-analysis among analysed works that quantifies the influence of individual determinants on the adoption of precision agriculture technologies. Based on the expected usefulness maximisation approach, Rahm and Huffman (1984) concludes that socioeconomic factors (specifically age, adopter's experience and financial situation) have a negligible effect on the adoption of precision agriculture technologies. According to this study, the key factors for adopting precision agriculture technologies are the economic benefit of the new technology (perceived profitability), technical literacy (defined as knowledge of work with computer technology), and the existence of advisory entities that support the dissemination of information or provide technical support. The profitability of new technologies as a fundamental determinant, which forms the basis for adoption considerations, is also highlighted by Lee et al. (2021). High up-front investment costs, along with potential benefits such as increased yields, optimized resource use, and subsequent reductions in operating costs, are crucial in achieving this profitability (John et al., 2023; Schimmelpfennig, 2016).

Other studies also highlight the importance of knowledge and awareness about new technologies in the adoption process, as well as the education of agricultural producers in general. This is supported by the findings of Cisternas et al. (2020), Yatribi (2020), da Silveira et al. (2021), Yarashynskaya and Prus (2022), and Osrof et al. (2023). Most studies emphasise the importance of technological determinants, particularly relative advantages and perceived usefulness (Degieter et al., 2023; Pathak et al., 2019; Sood et al., 2022; Yatribi, 2020). Therefore, it can be assumed that a lack of information about new technologies, leading to the inability to adequately evaluate the benefits of precision agriculture technologies, combined with low technical literacy, restricts their effective use and constitutes a significant barrier to their broader adoption by agricultural entities. At the same time, it is evident that political interventions can effectively influence these barriers. The relevance of this assumption and whether such interventions should be universally applied, as suggested by the findings of Tey and Brindal (2022), is the subject of further research.

Contribution of this study

The considerable growth in empirical studies on the drivers and barriers to adopting precision and digital agriculture highlights the need for a comprehensive synthesis and evaluation of the existing knowledge base. Such a synthesis offers researchers a detailed overview of the theoretical frameworks, methodological approaches, and data sources used in examining precision agriculture adoption. It also aids in identifying research gaps and delineating future research directions.

At a theoretical level, this synthesis facilitates the validation and development of theories related to adopting new technologies. On a practical level, it provides critical insights for developing effective policies and strategies to promote the adoption of precision and digital technologies. Ultimately, this synthesis serves as a crucial nexus between academic research and policy formulation, potentially catalyzing the widespread adoption of precision/digital agricultural technologies and fostering sustainable agricultural practices.

Previous studies synthesizing primary research on the factors influencing the adoption of precision farming technologies and digitalisation have predominantly employed a qualitative synthesis approach. Most reviews have focused separately on information and communication technology or precision agriculture rather than exploring the intersection of these two domains, with the exception of da Silveira et al. (2021). Some studies have incorporated both primary research and other reviews (e.g., Liu et al. (2021); da Silveira et al. (2021); Sood et al. (2022); Yarashynskaya and Prus (2022)).

In contrast, our study is based exclusively on primary studies that quantitatively analyse the determinants of adopting precision agriculture and digitalisation using logit or probit models. It incorporates the most recent findings, up to July 2024. Additionally, our approach complements both qualitative and quantitative synthesis and highlights the challenges associated with this research methodology. A literature review indicates that while systematic review methodologies are commonly used, meta-analyses examining the determinants of adoption are relatively rare. Among the existing meta-analyses, the study of Amoussouhoui et al. (2024) is notable; however, it specifically examines the correlation between age, gender, education, and income with the adoption rate of digital agricultural technologies.

In summary, our study offers three key contributions to the existing literature: (i) it is based exclusively on primary studies employing quantitative analysis; (ii) it is, to the best of our knowledge, the first to integrate both qualitative and quantitative synthesis, leveraging the synergy between these methods and highlighting associated methodological challenges; and (iii) it captures all relevant factors of adoption through meta-analysis, thereby avoiding potential biases from pre-selecting specific group of factors.

Materials and methods

The determinants of the adoption of precision agriculture technologies and digitalisation were defined through a systematic review and meta-analysis, followed the PRISMA protocol (Page et al., 2021; Shamseer et al., 2015). This systematic review identified key adoption determinants analysed in primary studies. The subsequent meta-analysis allowed the drawing of broad conclusions and the evaluation of factors influencing the adoption of precision and digital agriculture, thus determining the critical factors through quantitative data synthesis from these studies.

This study's systematic review and meta-analysis focused exclusively on primary studies that quantitatively examined the determinants of the adoption of precision agriculture technologies and digitalisation using regression analysis, specifically logit and probit models. Only studies whose results were fully published in English language scientific journals were included. Excluded from this research were conference papers, theoretical studies, review studies, and meta-analyses. Additionally, an exclusion criterion was applied to studies lacking relevant data, such as sample size and standard error of estimate.

The search for primary studies was conducted in the Web of Science (https://www.webofscience.com) and Scopus (https://www.scopus.com/) databases. The search used a sequence of four groups of English language keywords: (1) precision agriculture/smart agriculture/digital agriculture/agriculture 4.0/precision farming/smart farming/digital farming, (2) adoption/implementation/usage[6], (3) factors/drivers/barriers/determinants/motivations, (4) logit/probit/choice/regression. Terms within each keyword group were connected using "OR," and intergroup connections were made via "AND." The full syntax of the used SQL query is shown in Table 2.

From the obtained set of 733 studies, a unique DOI (Digital Object Identifier) was used to eliminate 273 duplicate studies. The remaining 460 studies were subjected to an assessment of their compliance with the required criteria, i.e., full-text publications in scientific journals presenting primary results in the English language, quantitatively examining driv-

[6] Abbreviation and the sign "*" were used to capture variants of these words.

Table 2 Syntax for finding primary studies

Databases	Syntax of the used SQL query
Web of Science $N=350$	TS = (((precision OR digital OR smart) NEAR/1 (agriculture OR farm*)) OR "agriculture 4.0") AND TS = (adopt* OR implement* OR use*) AND TS = ("factors" OR "drivers" OR "determinants" OR "barriers" OR motivat*) AND TS = ("choice" OR "probit" OR "logit" OR regress*) Refined by: Article (Document Types) and English (Languages) Indexes: Science Citation Index Expanded (SCI-EXPANDED) or Social Sciences Citation Index (SSCI) or Emerging Sources Citation Index (ESCI) (Web of Science Index)
SCOPUS $N=383$	TITLE-ABS-KEY (((precision OR digital OR smart) PRE/1 (agriculture OR farm*)) OR "agriculture 4.0") AND TITLE-ABS-KEY adopt* OR implement* OR use*) AND TITLE-ABS-KEY ("factors" OR "drivers" OR "barriers" OR "determinants" OR motivat*) AND TITLE-ABS-KEY ("choice" OR "logit" OR "probit" OR regress*) AND (LIMIT-TO (SRCTYPE, "j")) AND (LIMIT-TO (PUBSTAGE, "final")) AND (LIMIT-TO (DOCTYPE, "ar")) AND (LIMIT-TO (LANGUAGE, "English"))

Source: own processing

Note: The search was performed on July 23, 2024. N is the number of studies

ers and barriers to the adoption of precision agriculture and digitalisation, using the logit and probit model. The selection of the studies was performed in two steps, with the selection process carried out independently by two authors in accordance with the PRISMA protocol to ensure validity. The selection process first verified the compliance of each study with the selection criteria by evaluating name, abstract, and keywords. 410 studies were excluded in this step. The remaining studies were then subjected to an evaluation of the entire text. This selection excluded a further 28 studies. Thus, 22 studies were selected for the purpose of qualitative synthesis. The quality of presentation of primary study results, or more precisely, the availability of key data for meta-analysis, was also considered for quantitative analysis purposes. As a result of the absence of necessary data, a further 6 studies were excluded. The PRISMA flowchart in Fig. 1 illustrates the afore-mentioned material collection process for the research.

Data extraction

From the selected studies that met the selection criteria, three data categories were collected: 1) basic study characteristics, represented by: authors' names; year of publication; country of research; agricultural sector and precision/digital agriculture technology examined in the study; data (data source, type, period, sample size) and the methods used in the study; ii) factors influencing the adoption of precision/digital agriculture; and iii) the size of the effect, which refers to the quantitative estimate of how each factor impacts the adoption of precision agriculture and digitalisation. The impact of individual factors was described by their direction (positive/negative), magnitude, and significance[7] in influencing the adoption

[7] For every factor, the estimated coefficient and its standard error, or p-value, was recorded.

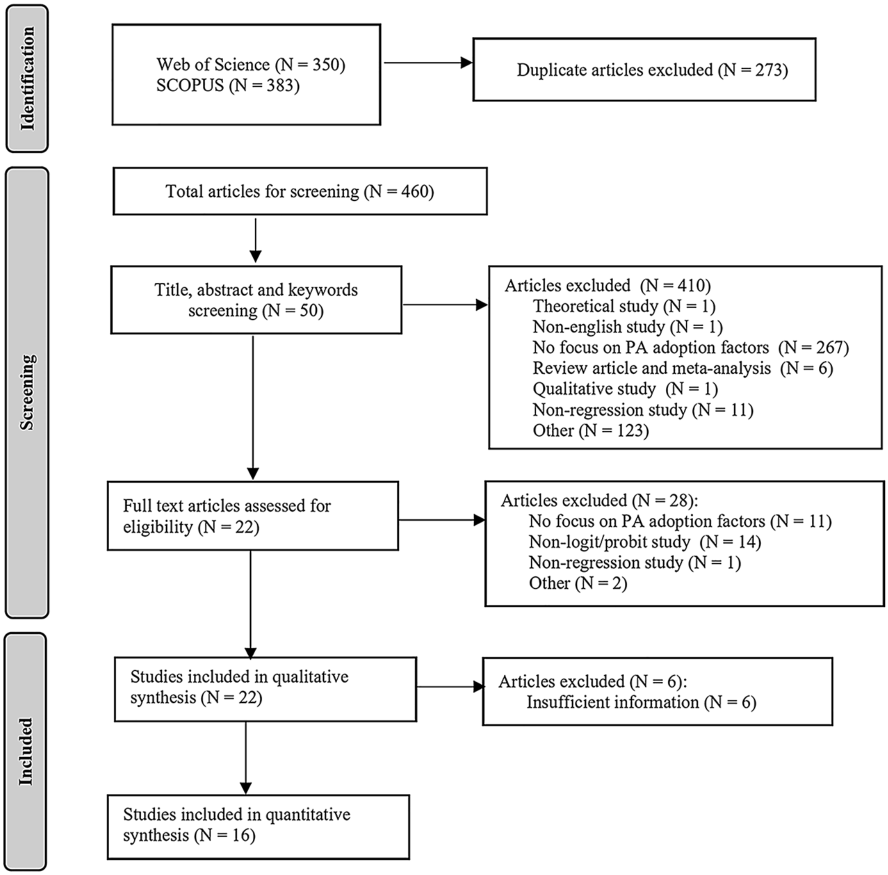

Fig. 1 PRISMA flowchart. Note: "Other" primarily represents studies that do not analyse precision agriculture, such as those focused on climate-smart agriculture practices (e.g., agroforestry and conservation tillage).

of precision/digital agriculture. Two authors independently verified the quality of the data extraction.

Qualitative synthesis

The results of the selected primary studies, whose basic characteristics are summarized in Table 3, were first subjected to a qualitative synthesis. This process involved defining individual factors for the adoption of precision agriculture technologies and digitalisation, categorising these factors and determining their importance based on the number of studies devoted to them, as well as the significance revealed in these studies.

The topicality of the adoption of precision agriculture is evident from Table 3. Most studies were published in the last five years. Only three studies were published before 2015, namely D'Antoni et al. (2012), Walton et al. (2010), Isgin et al. (2008). The regional focus of these studies was broad, whereby analyses working with data obtained by own questionnaire surveys in individual countries dominate. Only two studies (i.e., Barnes et al. (2019)

Table 3 Overview of studies included in the systematic review and meta-analysis

Author (year)	Country	Sector	Technology	Method	Data source	Analysed period	Sample size	RD
Bai et al. (2022)	Hungary	CP	Drones	Logistic regression	Primary data	2022	200	Y
Barnes et al. (2019)	Belgium, Germany, Greece, Netherlands, Great Britain	Wheat, potatoes, cotton	N/A	Multinomial + binomial random intercept model	Primary data	2016–2017	971	N
Blasch et al. (2021)	Austria	N/A	Fertiliser application technology	Multinomial + mixed + latent class logit model	Primary data	N/A	242	Y
Boyer et al. (2024)	USA	Beef	Cameras, drones, RFID, artificial insemination, checking for pregnancy, and computer software for production management	Logit model	Primary data	2022	201	N
D'Antoni et al. (2012)	USA	Cotton	Lightbar GPS	Multinomial logit model	Southern Cotton Precision Farming Survey	2009	469	Y
de Souza Filho et al. (2023)	Brazil	Sugar cane	Farm information system	Logit model	Primary data	2018–2019	131	Y
Franco et al. (2018)	India	Bananas	N/A	Logistic regression	Primary data	2012–2013	63	Y
Giua et al. (2022)	Italy	N/A	N/A	Zero-inflated Poisson model	Primary data	N/A	474	N
Groher et al. (2020a)	Switzerland	CP; vegetables, grapes, fruit, strawberries	Management assistance systems, Electronic measurement systems	Binary logistic regression	Primary data	N/A	608	Y
Groher et al. (2020b)	Switzerland	Dairy, beef, pork, poultry, eggs	ESMDs, ECs, EDPOs, robotics, electronic ear tags, smartphone usage.	Binary logistic regression	Primary data and Swiss Farm Structure Survey	2018	1497	Y

Table 3 (continued)

Author (year)	Country	Sector	Technology	Method	Data source	Analysed period	Sample size	RD
Isgin et al. (2008)	USA – Ohio	N/A	18 different technological components	Zero-inflated Poisson model, Zero-inflated negative binomial model	Ohio Farmland Lease and Precision Agriculture Survey	2003	491	N
Kabirigi et al. (2023)	Rwanda	Bananas	Smartphones	Binary logistic regression	Primary data	2018	690	Y
Kolady et al. (2021)	USA	Maize, soya, wheat	Embodied-knowledge technology, information-intensive technology	Probit model	Primary data	2017	166	Y
Michels et al. (2020)	Germany	N/A	Drones	Ordinal logit model	Primary data	2019	167	Y
Paustian and Theuvsen (2017)	Germany	CP	N/A	Binary logistic regression	Primary data	2014	227	Y
Piña et al. (2023)	USA	Beef	Big data	Binary probit model	Primary data	N/A	84	Y
Pivoto et al. (2019)	Brazil	Grains	N/A	Logit model, Poisson model	Primary data	N/A	119	Y
Tamirat et al. (2017)	Germany, Denmark	N/A	N/A	Binary logit model	Primary data	2008	206	Y
Vecchio et al. (2020)	Italy	N/A	N/A	Logit model	Primary data	N/A	174	Y
Walton et al. (2010)	USA	Cotton	Personal digital assistant	Probit model	Primary data	2005	827	N
Yue et al. (2023)	China	Apple	N/A	Probit model	Primary data	2019–2020	545	N
Zuo et al. (2021)	Australia	N/A	Drones	Probit model	Primary data	2015–2016	991	N

Source: own processing

Note: CP is crop production; RD indicates studies with sufficient data for quantitative synthesis (Y=yes, N=no); N/A refers to data not explicitly stated in the study; RFID denotes radio frequency identification; ESMDs denotes electronic sensors and measuring devices; ECs denotes electronic controls; EDPOs denotes electronic data-processing options. RD indicates studies with sufficient data for quantitative synthesis (Y=yes, N=no); N/A refers to data not explicitly stated in the study

and Tamirat et al. (2017)) processed data from multiple countries. Eight studies (specifically: Bai et al. (2022), Barnes et al. (2019), Blasch et al. (2021), Giua et al. (2022), Michels et al. (2020), Paustian and Theuvsen (2017), Tamirat et al. (2017), Vecchio et al. (2020)) analysed the adoption of precision agriculture in European Union member states, of which Germany was represented the most (four studies).

The analysed period was not explicitly defined in all the studies (e.g. Groher et al. (2020a); Pivoto et al. (2019)). Of the sixteen studies that provided this information, ten used data no older than 2015. When studies specified the agricultural sector in which they analysed the adoption of precision agriculture technologies, they predominantly focused on crop production or specific crops, with only three studies examining livestock production. Nearly half of the studies specified the technologies they focused on for adoption.

Among the analytical tools, the logit model using cross-sectional data dominated (e.g., Kabirigi et al. (2023)). The predominance of logit over the other type of discrete choice model – probit (e.g., Kolady et al. (2021) and Zuo et al. (2021)) – was probably the result of the higher demands of the probit model for fulfilling the assumption of normal distribution even on available data (Jose et al., 2020). The sample size in the analysis varied widely, from 63 to 1497 observations. Some reviewed studies worked with a relatively small sample size (in particular, Franco et al. (2018), Pivoto et al. (2019), and de Souza Filho et al. (2023)); nevertheless, this size was sufficient for estimating the parameters of the used analytical tools.

Quantitative synthesis

The quantitative synthesis was focused on effect size (ES), which states the strength of the factor in relation to the adoption of precision agriculture technologies and digitalisation. This study's observed effect sizes were based on regression coefficients estimated in primary studies. In the case of the logit model, the effect size of the i-th factor was represented directly by the logit coefficient of this factor (i.e., $ES_i = \widehat{\beta}_i$) and ES variability was expressed by the standard logit coefficient error ($SE(\widehat{\beta}_i)$). If the primary study applied a probit model, the effect size was obtained as 1.6 times the probit coefficient, and similarly ES variability was expressed as 1.6 times the standard probit coefficient error (Amemiya, 1981).

The total effect size for the i-th factor ($\overline{ES}_i$) was calculated as:

$$\overline{ES}_i = \frac{\sum_{i=1}^{I} \omega_{ij} ES_{ij}}{\sum_{i=1}^{I} \omega_{ij}} \tag{1}$$

where ES_{ij} is the effect size of the i-th factor in the j-th study and ω_{ij} is the weight assigned to the size of the effect of the i-th factor of the j-th study (Buck et al., 2022). This weight is quantified as:

$$\boldsymbol{\omega_{ij}} = \frac{1}{\boldsymbol{\sigma^2_{ij}} + \boldsymbol{\tau^2_i}} \tag{2}$$

where σ^2_{ij} is the variability of the ij-th effect size represented by ES variability (so-called within-study variance) (Baumgart-Getz et al., 2012) and τ^2_i is the variability of the true effect size of the i-th factor between studies (between-study variance) (Hamman et al., 2018), which describes the differences in the basic characteristics of the studies (DerSimonian & Laird, 1986). In this study, τ^2_i was estimated using the random effects (RE) model.

The random effects model estimate was performed in STATA 18 econometric software using the restricted maximum likelihood (REML) method. The extent of between-study variability was tested using a homogeneity test with the I^2 statistic, representing the proportion of between-study variability in total variability (Higgins & Thompson, 2002).

Heterogeneity analysis

In the case of high observable study heterogeneity, i.e., in the case of studies with different sets of explanatory variables or with varying methods of measuring similar variables, the synthesis of effect sizes from primary studies was expanded to include an analysis of differences in effect sizes between studies (Riley et al., 2011). For this, a meta-regression approach (MRA) was used, which includes predictor variables reflecting observable differences in the specification of primary study models (Aloe & Thompson, 2013) in the random effects model.

The RE-MRA model was expressed as (Ogundari & Bolarinwa, 2018):

$$ES_{ij} = \sum_{k=1}^{K} \gamma_k x_k + \epsilon_{ij} + \delta_{ij}, \tag{3}$$

where x_k are study characteristics (e.g., sample size, number of regressors, number of years, region, data type, the method used) and γ_k estimated parameters describing deviations from populational effect size (Aloe & Thompson, 2013), $\varepsilon_{ij} \sim (0, \sigma_{ij}^2)$, σ_{ij}^2 is the variability of the size of the effect of the i-th factor in the j-th study (within-study variance) and $\delta_{ij} \sim (0, \tau_i^2)$ is the difference between the average and the true effect specific for the given study (Veroniki et al., 2016). τ_i^2 is residual heterogeneity (Panityakul et al., 2013).

Publication bias

The estimated effect size may be skewed due to so-called publication bias, which results from prioritisatizing statistically significant results by authors, reviewers, or editors of scientific journals. This bias can lead to a situation where studies yielding relatively insignificant results remain unpublished (Rothstein et al., 2005). To verify the presence of this phenomenon, a funnel plot and a test of its asymmetry (funnel asymmetry test, FAT; Egger et al., 1997) were applied.

However, funnel plot asymmetry can also be caused by factors other than publication bias, namely by the presence of significant between-study heterogeneity. Therefore, this study used a modification of the funnel plot, enabling a differentiation between asymmetry as a consequence of publication bias and asymmetry as a result of other factors (so-called contour-enhanced funnel plot) (Peters et al., 2008). The effect of publication bias on meta-analysis results was evaluated using a nonparametric trim-and-fill method (Duval & Tweedie, 2000).

Results and discussion

A qualitative synthesis identified almost 70 variables explaining the adoption of precision agriculture technologies and digitalisation. However, not all these variables represent unique determinants, i.e., drivers of or barriers to adopting precision/digital agriculture. Some cases involve the same determinant, but in a different expression; see Table 4.

Although the differences between the variables used in these studies reduce the possibility of generalising conclusions about their effect on the adoption of precision agriculture technologies and digitalisation, it can be clearly stated that the main emphasis in currently available literature is focused on socioeconomic factors. According to the qualitative synthesis, the adoption of precision agriculture technologies and digitalisation is significantly influenced by both farmers` characteristics, including their age, gender, education, and farm attributes. This finding aligns with the results of previous studies (Osrof et al., 2023; Amoussouhoui et al., 2024). For example, Osrof et al. (2023) identified education, farm size, perceived benefits, and access to information about technology as the main drivers of adoption. Conversely, they identified aging, financial cost, perceived complexity, and lack of well-developed infrastructure as the most frequent barriers to adoption.

The most frequently represented determinant in the evaluated studies is farm size, encompassing various metrics such as farmland area, number of workers, and livestock units, which are considered in a total of 25 studies. The predominant effect of this determinant on the adoption of precision agriculture technologies and digitalisation is positive and statistically significant at 5% level. Other most frequently occurring factors are age and education. The effect of these determinants is evaluated identically in a total of 17 studies. In the case of age, the predominant effect on the adoption of precision agriculture and digitalisation is negative, while in the case of education, it is positive. However, the impact of both of these determinants is not clearly significant. In the case of age, a significant effect at a significance level of 5% was revealed in only eight studies. A significant positive effect of education on the adoption of precision agriculture technologies and digitalisation was observed in analyses focusing on specific technologies, such as smartphones and farm management information systems (FMIS). Additionally, participation in training sessions and workshops, which enhances technical literacy and serves as a key information source, also positively influences the adoption of precision agriculture technologies and digitalisation.

The significant role of economies of scale can be identified in economic factors, which increase farmers‘ ability to acquire capital-intensive technologies such as those used in precision agriculture. 14 studies included income, sales, and yields among adoption determinants. Most of these studies revealed a positive effect, including income from nonagricultural activity, which is a source of external capital. Although farmers‘ perception of profitability is considered the most important motivating factor in adopting precision agriculture and digitalisation (Lowenberg-DeBoer & Erickson, 2019), available studies do not analyse this phenomenon very much. Only two studies address perceived profitability and expected return on investment in precision agriculture technologies. Both, however, find a positive statistically significant relationship between this perceived economic benefit and the adoption of precision agriculture technologies.

Interestingly, research by Schimmelpfennig (2016) provides concrete evidence supporting these perceptions of profitability. Schimmelpfennig’s study addresses the impact of adopting selected precision agriculture technologies (GPS-based mapping systems, guid-

Table 4 Determinants of adoption of precision/digital agriculture

Determinant	Description	NV/NZ	Predominant direction of action
Age	Discrete variable, number of years	8/14	-
Age	Categorial variable	0/3	-
Age	Binary variable, D = 1 for < 50 years	0/1	+
Gender	Binary variable, D = 1 pro men	1/5	+
Gender	Binary variable, D = 1 pro female	1/1	-
Marital status	Binary variable, D = 1 for marriage	0/1	+
Education	Categorial variable, or Binary variable for university education (D = 1)	2/12	+
Education	Binary variable, D = 1 for short training after primary school	0/1	+
Education	Binary variable, D = 1 for longer education, including university	0/1	+
Education	Discrete variable, number of years	1/4	+
Education	Binary variable, D = 1 for less than 1 years‘ study	0/1	-
Agricultural education	Binary variable, D = 1 for PA literacy	1/2	+
Technological literacy	Binary variable, D = 1 for university education in an agricultural field	1/1	+
Experience	Discrete variable, number of years	1/4	-
Agricultural experience (CP)	Binary variable, D = 1 for < 5 years	0/1	+
Agricultural experience (CP)	Binary variable, D = 1 for 16–20 years	1/1	+
Size of enterprise	Continuous variable, ha of agricultural land or ha of arable land	10/14	+
Size of enterprise	Binary variable, D = 1 for < 250 ha	2/3	-
Size of enterprise	Binary variable, D = 1 for > 750 ha	1/3	+
Size of enterprise	Categorial variable	1/1	+
Labour force	Discrete variable, number of workers	0/1	+
Herd size	Discrete variable, total number of livestock units	2/2	+
Herd size	categorical variable representing number of head	1/1	+
Use of computer	Binary variable, D = 1 for IT use	2/4	+
Income	Binary variable, D = 1 for income exceeding a certain threshold	0/2	+
Household income	Binary variable, D = 1 pro $50,000–499,999	0/1	-
Income	Categorial variable	0/2	+
Income	Continuous variable, monetary units	1/1	+
Income from nonagricultural activity	Continuous variable, percentage of income from nonagricultural activity	1/1	+
Income from nonagricultural activity	Binary variable, D = 1 for yes	1/2	+
Income from agricultural activity	Categorial variable	0/1	-/+
Sales	Continuous variable, monetary units	1/2	+
Revenue	Continuous variable, yield from ha of land	0/1	+
Indebtedness	Continuous variable, ratio of total debt to total assets	1/1	+
Debt	Continuous variable, monetary units	1/1	+
Obtainment of credit	Binary variable, D = 1 for yes	1/1	+

Table 4 (continued)

Determinant	Description	NV/NZ	Predominant direction of action
Specialisation	Binary variable, D = 1 for individual production orientations	4/7	-/+
Specialisation	Continuous variable, proportion of arable land in agricultural land	1/1	+
Specialisation	Binary variable, D = 1 for proportion of income from a certain crop exceeding 60%	0/1	+
Farm plan	Binary variable, D = 1 for yes	1/1	+
Extent of PA adoption	scale of 0–4, where 0 = no PA technologies, 4 = 4 other PA technologies	1/1	+
Hierarchical integration	Categorial variable	1/2	+
Membership of cooperatives and associations	Binary variable, D = 1 for yes	1/2	-/+
Organic farming	Binary variable, D = 1 for organic farming	0/2	-/+
Work intensity	Continuous variable, days/ha	1/1	+
Land ownership	Continuous variable, proportion of land owned by farmers in total farm area	1/1	+
Soil quality	Continuous variable, the ratio of farm's hectare yields to the national average	1/1	+
Soil productivity	Binary variable, D = 1 for income/ha >$420	0/1	+
Land value	Continuous variable, monetary units	0/1	-
Irrigation systems	Continuous variable, percentual representation of irrigation systems	0/1	+
Technology characteristics (expected increase of yield)	Categorial variable	0/1	+
Technology characteristics (expected improvement of water quality)	Categorial variable	1/1	+
Technology characteristics (expected fertiliser reduction)	Categorial variable	1/1	+
Technology characteristics (degree of automation)	Categorial variable	0/1	-/+
Technology characteristics (acquisition costs)	Continuous variable, monetary units	1/1	-
Technology characteristics (availability of free assistance)	Binary variable, D = 1 for yes	1/1	+
Technology characteristics (perceived complexity)	Likert scale	1/1	-
Information sources	Binary variable, D = 1 for consultants, suppliers, other agricultural enterprises	0/1	-/+
Access to digital information	Binary variable, D = 1 for using the internet	1/2	+
Participation in workshops and training sessions	Binary variable, D = 1 for yes	1/2	-/+
Use of technical assistance	Binary variable, D = 1 for yes	0/1	+
Frequency of technical assistance	Continuous variable, number of days in the year	0/1	-
Tendency to use technologies	Categorial variable	1/1	+
Uncertainty of future outputs	Binary variable, D = 1 for yes	0/1	-

Table 4 (continued)

Determinant	Description	NV/NZ	Predominant direction of action
Expected return on investment	Binary variable, D = 1 for positive statement regarding expected return of investment	1/1	+
Information about PA	Binary variable, D = 1 for sufficient	0/1	-
Information about PA	Binary variable, D = 1 for insufficient or don't know	1/1	-
Perception of PA	Binary variable, D = 1 for positive perception of PA	0/1	+
Perception of PA	Binary variable, D = 1 for don't know or missing response	0/1	-
Risk orientation	Likert scale	1/2	-
Insurance	Binary variable, D = 1 for yes	1/1	+
Trust in the privacy of farm data	Likert scale	1/1	+
Innovation	Likert scale	0/1	+
Government support	Discrete variable, number of visits to enterprise	0/1	-
Government subsidies	Binary variable, D = 1 for yes	0/1	+
Government regulations	Binary variable, D = 1 for environmental regulations	0/1	+
Management services	Binary variable, D = 1 for use	1/1	+
Profitability index	Likert scale, perceptions of profitability benefits from PA	1/1	+
Environmental index	Likert scale, perceptions of environmental benefits from PA	1/1	-

Source: own processing

Note: NV indicates the number of studies that reveal a statistically significant effect of the given adoption determinant, i.e., at a significance level of 5%; NZ is the number of studies that evaluate the given adoption determinant; PA is precision agriculture

ance or auto-steer systems, and variable-rate technology) on profitability. The findings indicate that these technologies have a small but positive impact on both net returns and operating profit for average-sized US corn farms. Notably, GPS mapping shows the largest estimated impact, increasing operating profit by almost 3% and net returns by nearly 2%. Furthermore, Schimmelpfennig observes that, on average, the operating profit of precision agriculture adopters was higher than that of non-adopters.

Other factors worth noting are farmers‘ gender and computer use. Results show that men are traditionally more open to adopting new technologies. Two of the six studies that analyse this factor report a statistically significant effect of gender. Computer use is a significant factor in this process, especially in developing countries with limited access to information technology. Despite its recognized importance, the positive effect of computer use on this process is not statistically significant at the 5% level.

Studies further indicate that farmers with a comprehensive understanding of precision agriculture and a risk-taking propensity are likelier to adopt these technologies. Additionally, the dissemination of relevant information plays a crucial role in this adoption process. Information is transmitted through various channels, including peer farmers, technology companies, research centers, private consultants, and agricultural unions. Enhanced access to information via extension services correlates with a greater likelihood of farmers innovating and adopting precision farming technologies. Increased awareness of the advantages of precision agriculture and digitalisation can further drive the adoption of these practices among farmers.

The effect of other determinants on the adoption of precision/digital agriculture cannot be generalised, as in most cases it is examined in one or only a few publications. A limiting factor of the entire synthesis is also the non-uniform nature of the considered determinants. For instance, specialization is variously defined as a continuous, categorical, or even binary variable. Furthermore, studies focus on diverse production orientations, which complicates comparisons further.

An objective comparison of the direction and magnitude of the effect of individual factors on the adoption of precision agriculture technologies and digitalisation is made possible by quantitative synthesis. However, significant heterogeneity in the reviewed studies severely limits this comparison to only a few factors from a handful of studies. For example, the variables of experience in agricultural production and age are expressed in some studies by several years, while in others, they are dummy variables for time (age) categories. Education is also expressed in some studies over several years, while in others, it is expressed by dummy variables for the highest achieved education or only a dummy variable for third-level education. Size is expressed in some studies directly by hectares, while in others by categories, which, however, are not defined in the same way; a small farm is defined in some studies as up to 100 ha, while in other studies, it is up to 1,000 ha. Even when studies measure a specific determinant consistently, quantitative synthesis is hindered by disparities in results presentation, with some studies reporting estimated parameter values and others presenting marginal effects. Gender exemplifies this issue. The aforementioned fact, which reduces the number of studies included in the quantitative synthesis, makes it impossible to fully address the existing research gap.

Table 5; Fig. 2 in the appendix present total effect sizes for comparable studies' most frequently represented adoption determinants. At a 5% level, the statistically significant effect size is revealed for farm size expressed in hectares and third-level education. This indicates that large and capital-strong farms are more likely to implement new technologies in their production practice. The results also emphasise the positive effect of education on adopting precision agriculture and digitalisation. The effect size for farmer age is not statistically significant.

The I^2 values reveal that high between-study heterogeneity exists in the analysed set regarding size and age. In the case of age, this heterogeneity is explained by the characteristics of the reviewed studies; see Tables 6, 7 and 8 in the appendix. In the case of size, the observed differences between the studies are insufficient to explain between-study heterogeneity, which may indicate the existence of publication bias. The form of the funnel plots also evidences this; see Figs. 3, 4 and 5 in the appendix, the test of their asymmetry, and the estimate of the total effect size using the trim-and-fill method, which is presented in

Table 5 Results of the meta-analysis of determinants of the adoption of precision agriculture and digitalisation

Determinant	Effect size	*p*-value	Confidence interval (95%)		I^2 (%)
Education	0.510	0.009	0.127	0.893	26.51
Size	0.497	0.027	0.058	0.936	100
Age	− 0.009	0.419	− 0.030	0.013	75.52

Source: own processing

Table 9 in the appendix. According to the results of these statistical tools, the size variable shows publication bias, whereby it cannot be ruled out that the missing studies might reveal a statistically insignificant effect of the size of the enterprise on the adoption of precision agriculture technologies and digitalisation.

Analysis of between-study heterogeneity reveals that the influence of individual determinants on precision farming adoption appears to be geographically specific. Our qualitative synthesis encompassed a diverse sample of studies with a notable geographical distribution: 45% of the studies investigated precision farming adoption in European states (including Switzerland and Great Britain). In comparison, 22% focused on adoption patterns in the United States. Specifically, Bai et al. (2022); Barnes et al. (2019); Blasch et al. (2021); Groher et al. (2020a); Michels et al. (2020); Paustian and Theuvsen (2017); Tamirat et al. (2017); Vecchio et al. (2020) focus on the issue of precision/digital agriculture adoption and its determinants in a European context. However, even based on these studies, it is difficult to draw definite conclusions and recommendations for European states since each of the studies focuses on a different sector, and it is evident that every sector has its own requirements and specifics.

Our study's findings align with the general consensus in the existing literature on the adoption of precision agriculture and digitalisation, yet they also highlight some unique nuances. For instance, the significant role of farm size and education in the adoption process is well-documented, corroborating results from previous studies such as Barnes et al. (2019) and Yarashynskaya and Prus (2022). Due to their greater access to capital and economies of scale, large farms are more likely to adopt precision agriculture technologies, as evidenced by a statistically significant effect size of 0.497 for farm size and 0.510 for education at a 5% significance level. This is consistent with findings by Blasch et al. (2022) and Troiano et al. (2023), who also noted the positive impact of these factors on technology adoption. Furthermore, our results on the positive influence of younger, more educated farmers adopting new technologies align with Yarashynskaya and Prus (2022), indicating that these demographic factors play a crucial role in the diffusion of digital farming practices.

However, our study reveals significant heterogeneity in the impact of these determinants across different geographic regions and agricultural sectors, which needs to be more frequently emphasized in the existing literature. For example, while the effect of age on adoption was not statistically significant in our overall analysis, it exhibited substantial variability when examined within specific contexts, such as different regions or crop types. This heterogeneity underscores the need for localized studies to understand the adoption dynamics in various settings better. The funnel plots and tests of asymmetry for size and education determinants indicated potential publication bias, further complicating the synthesis of these findings. Studies like those by Bai et al. (2022) and Michels et al. (2020) also emphasize that sector-specific and region-specific factors significantly influence the adoption of precision agriculture technologies, which should be considered when formulating policies and strategies. This calls for a nuanced approach to policy-making considering the unique characteristics of different agricultural sectors and regions to support the effective adoption of precision agriculture technologies and digitalisation.

Moreover, our analysis highlights the critical role of perceived economic benefits in adopting precision agriculture technologies, a factor consistently identified in the literature. Studies by Tey and Brindal (2022) and Lee et al. (2021) underscore the importance of profitability as a key driver for technology adoption. Our findings echo this sentiment, with perceived profitability showing a positive and statistically significant relationship with adopting precision agriculture technologies. This suggests that farmers are more inclined to adopt these technologies if they anticipate significant cost savings and yield improvements, reinforcing the need for clear communication of the economic benefits associated with precision agriculture. Additionally, the presence of advisory services and technical support, as noted by Cisternas et al. (2020) and Sood et al. (2022), emerged as crucial facilitators in our study, indicating that support structures play a vital role in bridging the knowledge gap and encouraging adoption.

However, our study also identifies several barriers that hinder the widespread adoption of precision agriculture technologies, which should be more prominently discussed in existing literature. One such barrier is the high initial investment cost, which remains a significant deterrent for small and medium-sized farms. This finding aligns with studies by Pathak et al. (2019) and John et al. (2023), which also noted the financial constraints faced by smaller agricultural enterprises. Furthermore, these technologies' complexity and perceived ease of use were frequently cited as barriers, corroborating findings by Degieter et al. (2023) and Osrof et al. (2023). These technological barriers highlight the need for user-friendly, cost-effective solutions to facilitate broader adoption. Additionally, our study found that institutional factors such as access to credit and government support play a pivotal role in adoption, consistent with the conclusions of Yarashynskaya and Prus (2022) and Blasch et al. (2022). These findings suggest that policy interventions to reduce financial barriers and provide technical support could significantly enhance the adoption rates of precision agriculture technologies.

Conclusion and implications

Precision farming adoption and digitalisation bring innovations into agriculture worldwide. Each region faces new challenges that cover regional characteristics and limitations. European agricultural producers are currently facing a number of challenges, among which the European Green Deal, the war in Ukraine and labour shortages can be highlighted. These challenges intensify the pressure on agricultural productivity, i.e., ensuring high-quality and affordable foodstuffs and agricultural raw materials with minimal negative impacts on the environment. Precision agriculture and digitalisation are two of the ways to meet the demands placed on 21st century agriculture.

Although these technologies have numerous social, economic, and environmental benefits, their diffusion and adoption by agricultural producers still need to be improved and faster. Therefore, understanding the drivers of and barriers to the adoption of precision agriculture technologies and digitalisation is important from a microeconomic and political perspective. It allows policy-makers to formulate valuable and effective support for the diffusion and adoption of these technologies by focusing on key factors of adopting precision agriculture and digitalisation. Based on this motivation, the presented study aimed to identify factors that influence the adoption of precision/digital agriculture technologies and determine their importance and significance in adopting precision agriculture and digitalisation.

Based on both qualitative and quantitative synthesis, the adoption of precision agriculture technologies is significantly influenced by farmer characteristics—such as age and education—as well as the parameters of the agricultural enterprise. Most studies, irrespective of sector or geographical region, indicate that the willingness to adopt new technologies declines with the farmer's age. Conversely, younger and well-educated farmers exhibit a greater propensity for adopting precision agriculture and digitalisation, highlighting the positive role of human capital in technology diffusion. Moreover, larger and more capital-intensive farms are more likely to implement new agricultural innovations, likely due to economies of scale and greater financial resources. While these factors are crucial and should be considered when specifying models for precision agriculture and digitalisation adoption, other factors are also important. Specifically, perceived economic benefits, such as profitability and expected return on investment, are key motivators for technology adoption. This reinforces the importance of effectively communicating the economic advantages of precision agriculture to farmers. Additionally, the availability of advisory services and technical support structures emerged as crucial facilitators, suggesting that policies and interventions aimed at bridging knowledge gaps and providing technical assistance could significantly enhance the uptake of these modern agricultural technologies.

The synthesis of research on precision/digital agriculture adoption faces several significant challenges. The effects of many determinants cannot be generalized due to limited coverage in existing literature, with most being examined in only one or a few publications. Furthermore, the non-uniform nature of considered determinants complicates comparisons. Even when studies measure a specific determinant consistently, quantitative synthesis is hindered by disparities in results presentation, with some studies reporting estimated parameter values and others presenting marginal effects. The geographical stratification suggests that region-specific factors, such as variations in agricultural policies and technological infrastructure influence adoption behaviour. The observed heterogeneity underscores the importance of considering local contexts when generalizing findings related to precision farming adoption across different geographic regions. Future research could be oriented towards a comprehensive synthesis of findings from studies focused on specific agricultural sectors and individual countries, allowing for a more nuanced understanding of precision agriculture adoption patterns within particular contexts. However, the successful execution of such meta-analytical approaches necessitates a standardization of methodologies across primary studies, specifically in terms of the operationalization and measurement of adoption determinants and the presentation and reporting of research outcomes. This standardization is crucial for ensuring the comparability and aggregation of results, thereby enhancing the validity and reliability of subsequent meta-analyses. By addressing these methodological considerations, researchers can facilitate more robust cross-study comparisons and derive more generalizable insights into the sector-specific and country-specific dynamics of precision agriculture adoption, ultimately contributing to a more cohesive and nuanced body of knowledge in this field.

Appendix

See Figure 2, Tables 6, 7, 8, Figures 3, 4, 5 and Table 9.

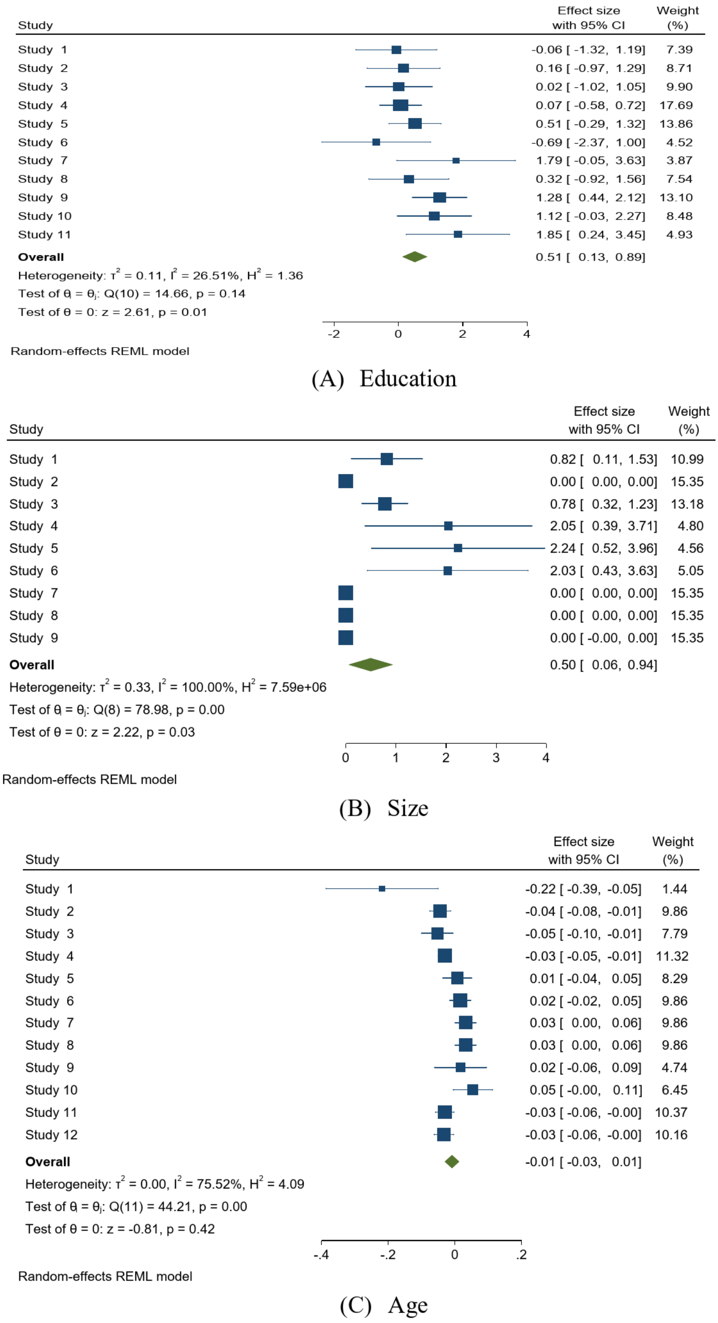

Fig. 2 Forestplot

Table 6 Estimation of a meta-regression model for age

Age	Coefficient	Statistical error	*p*-value
EU	− 0.062	0.012	0.000
USA	0.003	0.012	0.808
Sample size	0.001	0.000	0.000
Panel	− 0.240	0.049	0.000
Const.	− 0.070	0.025	0.006
Tau^2	1.9e-07		
I^2_res (%)	0.05		
R^2 (%)	99.98		
Wald chi^2[4]	40.17		0.000
Residual heterogeneity test Q_chi^2[7]	4.02		0.778

Source: own processing

Table 7 Estimation of a meta-regression model for education

Education	Coefficient	Statistical error	*p*-value
EU	1.233	0.505	0.015
USA	− 0.224	0.407	0.581
Sample size	− 0.012	0.008	0.107
Const.	2.324	1.219	0.057
Tau^2	3.7e-08		
I^2_res (%)	0		
R^2 (%)	100		
Wald chi^2[3]	8.28		0.041
Residual heterogeneity test Q_chi^2[7]	6.38		0.496

Source: own processing

Table 8 Estimation of a meta-regression model for size

Size	Coefficient	Statistical error	*p*-value
EU	0.098	0.028	0.000
USA	2.185	0.489	0.000
Sample size	− 0.014	0.004	0.000
Const.	2.273	0.628	0.000
Tau^2	2.9e-06		
I^2_res (%)	81.85		
R^2 (%)	100		
Wald chi^2 [3]	32.12		0.000
Residual heterogeneity test Q_chi^2[5]	37.24		0.000

Source: own processing

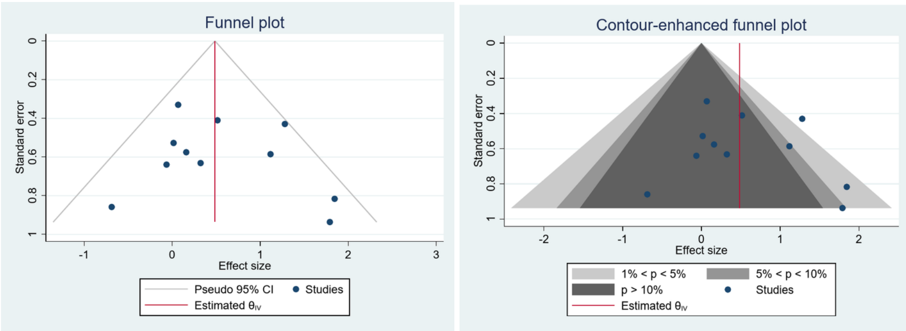

Fig. 3 Funnel plots for education. Note: A funnel plot is quite asymmetric with a skew towards the right side. This asymmetry is particularly evident for studies with larger standard errors. In the case of a study lying outside the gray areas, the null hypothesis of a statistically insignificant effect is rejected at the 1% significance level. This asymmetry in the funnel plot could indicate potential publication bias. However, that asymmetry can also be caused by heterogeneity

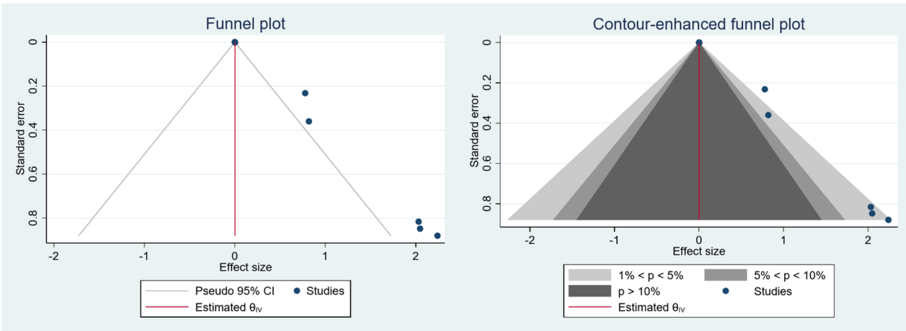

Fig. 4 Funnel plots for size. Note: The funnel plot is asymmetric. In the non-significant area of the enhanced plot, small studies are missing, which indicates the presence of publication bias

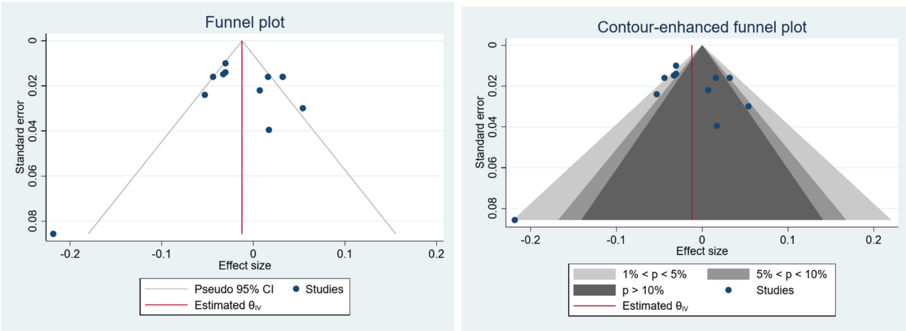

Fig. 5 Funnel plots for age. Note: The funnel plot is quite symmetric. It is evident that there is a significant outlier study with a notably negative but less precise effect (high SE value). For the two studies lying outside the shaded areas, the null hypothesis of no statistically significant effect is rejected at the 1% significance level. These studies can be characterized by a low value of standard error, i.e., high precision. Hypothetically missing studies, i.e., studies that would make the plot symmetrical around the red vertical line, would fall into the area with a p-value greater than 10% and a negative effect size. That is, the slight asymmetry of the plot is more likely attributed to heterogeneity rather than publication bias

Table 9 Results of the sensitivity analysis

Determinant	Effect size	Confidence interval (95%)		FAT (*p*-value)
Age	− 0.009	− 0.030	0.013	0.556
Education	0.442	0.051	0.832	0.508
Size	0.209	− 0.442	0.861	0.000

Source: own processing

Note: According to the funnel plot asymmetry test, the null hypothesis of no effect from small studies cannot be rejected at the 5% significance level for age and education, indicating that the results are not affected by publication bias. Conversely, for size, the null hypothesis of no effect from small studies is rejected at the 5% significance level, even when the test is extended to include moderators

Acknowledgements The results are part of Project No. QK 23020058, “Precision Agriculture and Digitization in the Czech Republic”, supported by the Ministry of Agriculture of the Czech Republic under the ZEME program.

Funding Funding was provided by the Ministry of Agriculture of the Czech Republic, under the ZEME program, Project No. QK 23020058, “Precision Agriculture and Digitization in the Czech Republic”.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Aloe, A., & Thompson, C. (2013). The Synthesis of Partial Effect Sizes. *Journal of the Society for Social Work and Research*, *4*(4), 390–405. https://doi.org/10.5243/jsswr.2013.24

Amemiya, T. (1981). Qualitative Response Models: A Survey. *Journal of Economic Literature*, *19*(4), 1483–1536.

Amoussouhoui, R., Arouna, A., Ruzzante, S., & Banout, J. (2024). Adoption of ICT4D and its determinants: A systematic review and meta-analysis. *Heliyon*, *10*(9). https://doi.org/10.1016/j.heliyon.2024.e30210

Bai, A., Kovách, I., Czibere, I., Megyesi, B., & Balogh, P. (2022). Examining the Adoption of Drones and Categorisation of Precision Elements among Hungarian Precision Farmers Using a Trans-Theoretical Model. *Drones*, *6*(8). https://doi.org/10.3390/drones6080200

Barnes, A., Soto, I., Eory, V., Beck, B., Balafoutis, A., Sánchez, B., Vangeyte, J., Fountas, S., van der Wal, T., & Gómez-Barbero, M. (2019). Exploring the adoption of precision agricultural technologies: A cross-regional study of EU farmers. *Land Use Policy*, *80*, 163–174. https://doi.org/10.1016/j.landusepol.2018.10.004

Baumgart-Getz, A., Prokopy, L., & Floress, K. (2012). Why farmers adopt best management practice in the United States: A meta-analysis of the adoption literature. *Journal of Environmental Management*, *96*(1), 17–25. https://doi.org/10.1016/j.jenvman.2011.10.006

Blasch, J., Vuolo, F., Essl, L., & van der Kroon, B. (2021). Drivers and Barriers Influencing the Willingness to Adopt Technologies for Variable Rate Application of Fertiliser in Lower Austria. *Agronomy*, *11*(10). https://doi.org/10.3390/agronomy11101965

Blasch, J., van der Kroon, B., van Beukering, P., Munster, R., Fabiani, S., Nino, P., & Vanino, S. (2022). Farmer preferences for adopting precision farming technologies: a case study from Italy. *European Review of Agricultural Economics*, *49*(1), 33–81. https://doi.org/10.1093/erae/jbaa031

Boyer, C., Cavasos, K., Greig, J., & Schexnayder, S. (2024). Influence of risk and trust on beef producers' use of precision livestock farming. *Computers and Electronics in Agriculture*, *218*. https://doi.org/10.1016/j.compag.2024.108641

Buck, R., Fieberg, J., & Larkin, D. (2022). The use of weighted averages of Hedges' d in meta-analysis: Is it worth it? *Methods in Ecology and Evolution*, *13*(5), 1093–1105. https://doi.org/10.1111/2041-210X.13818

Bulut, C., & Wu, P. (2023). More than two decades of research on IoT in agriculture: a systematic literature review. *Internet Research*, *34*(3), 994–1016. https://doi.org/10.1108/INTR-07-2022-0559

Cisternas, I., Velásquez, I., Caro, A., & Rodríguez, A. (2020). Systematic literature review of implementations of precision agriculture. *Computers and Electronics in Agriculture*, 176. https://doi.org/10.1016/j.compag.2020.105626

D'Antoni, J., Mishra, A., & Joo, H. (2012). Farmers' perception of precision technology: The case of autosteer adoption by cotton farmers. *Computers and Electronics in Agriculture*, *87*, 121–128. https://doi.org/10.1016/j.compag.2012.05.017

da Silveira, F., Lermen, F., & Amaral, F. (2021). An overview of agriculture 4.0 development: Systematic review of descriptions, technologies, barriers, advantages, and disadvantages. *Computers and Electronics in Agriculture*, 189. https://doi.org/10.1016/j.compag.2021.106405

Davis, F., Bogozzi, R., & Warshaw, P. (1989). User acceptance of computer technology: A comparison of two theoretical models. *Management Science*, *35*, 982–1003.

de Souza Filho, H., Vinholis, M., Carrer, M., & Mozambani, C. (2023). Adoption of farm management information systems (FMIS): The case of Brazilian sugarcane farmers. *Information Development*. https://doi.org/10.1177/02666669231177864

Degieter, M., De Steur, H., Tran, D., Gellynck, X., & Schouteten, J. (2023). Farmers' acceptance of robotics and unmanned aerial vehicles: A systematic review. *Agronomy Journal*, *115*(5), 2159–2173. https://doi.org/10.1002/agj2.21427

DerSimonian, R., & Laird, N. (1986). Meta-analysis in clinical trials. *Controlled Clinical Trials*, *7*(3), 177–188. https://doi.org/10.1016/0197-2456(86)90046-2

Duval, S., & Tweedie, R. (2000). A Nonparametric Trim and Fill Method of Accounting for Publication Bias in Meta-Analysis. *Journal of the American Statistical Association*, *95*(449), 89–98. https://doi.org/10.1080/01621459.2000.10473905

Egger, M., Smith, G., Schneider, M., & Minder, C. (1997). Bias in meta-analysis detected by a simple, graphical test. *Bmj*, *315*(7109), 629–634. https://doi.org/10.1136/bmj.315.7109.629

Franco, D., Singh, D., & Praveen, K. (2018). Evaluation of Adoption of Precision Farming and its Profitability in Banana Crop. *Indian Journal of Economics and Development*, *14*(2). https://doi.org/10.5958/2322-0430.2018.00124.5

Giua, C., Materia, V., & Camanzi, L. (2022). Smart farming technologies adoption: Which factors play a role in the digital transition? *Technology in Society*, *68*. https://doi.org/10.1016/j.techsoc.2022.101869

Grennhalgh, T., Robert, G., Macfarlane, F., Bate, P., & Kyriakidou, O. (2004). Diffusion of Innovations in Service Organizations: Systematic Review and Recommendations. *The Milbank Quarterly*, *82*(4), 581–629. https://doi.org/10.1111/j.0887-378X.2004.00325.x

Groher, T., Heitkämper, K., Walter, A., Liebisch, F., & Umstätter, C. (2020a). Status quo of adoption of precision agriculture enabling technologies in Swiss plant production. *Precision Agriculture*, *21*(6), 1327–1350. https://doi.org/10.1007/s11119-020-09723-5

Groher, T., Heitkämper, K., & Umstätter, C. (2020b). Digital technology adoption in livestock production with a special focus on ruminant farming. *Animal*, *14*(11), 2404–2413. https://doi.org/10.1017/S1751731120001391

Hamman, E., Pappalardo, P., Bence, J., Peacor, S., & Osenberg, C. (2018). Bias in meta-analyses using Hedges' d. *Ecosphere*, *9*(9). https://doi.org/10.1002/ecs2.2419

Higgins, J., & Thompson, S. (2002). Quantifying heterogeneity in a meta-analysis. *Statistics in Medicine*, *21*(11), 1539–1558. https://doi.org/10.1002/sim.1186

Isgin, T., Bilgic, A., Forster, D., & Batte, M. (2008). Using count data models to determine the factors affecting farmers' quantity decisions of precision farming technology adoption. *Computers and Electronics in Agriculture*, *62*(2), 231–242. https://doi.org/10.1016/j.compag.2008.01.004

John, D., Hussin, N., Shahibi, M., Ahmad, M., Hashim, H., & Ametefe, D. (2023). A systematic review on the factors governing precision agriculture adoption among small-scale farmers. *Outlook on Agriculture*, *52*(4), 469–485. https://doi.org/10.1177/00307270231205640

Jose, A., Philip, M., Prasanna, L., & Manjula, M. (2020). Comparison of Probit and Logistic Regression Models in the Analysis of Dichotomous Outcomes. *Current Research in Biostatistics*, *10*(1), 1–19. https://doi.org/10.3844/amjbsp.2020.1.19

Kabirigi, M., Sekabira, H., Sun, Z., & Hermans, F. (2023). The use of mobile phones and the heterogeneity of banana farmers in Rwanda. *Environment Development and Sustainability*, *25*(6), 5315–5335. https://doi.org/10.1007/s10668-022-02268-9

Khanal, A., Mishra, A., Lambert, D., & Paudel, K. (2019). Modeling post adoption decision in precision agriculture: A Bayesian approach. *Computers and Electronics in Agriculture*, *162*, 466–474. https://doi.org/10.1016/j.compag.2019.04.025

Klerkx, L., Jakku, E., & Labarthe, P. (2019). A review of social science on digital agriculture, smart farming and agriculture 4.0: New contributions and a future research agenda. *NJAS: Wageningen Journal of Life Sciences*, *90–91*(1), 1–16. https://doi.org/10.1016/j.njas.2019.100315

Kolady, D., Van der Sluis, E., Uddin, M., & Deutz, A. (2021). Determinants of adoption and adoption intensity of precision agriculture technologies: evidence from South Dakota. *Precision Agriculture*, *22*(3), 689–710. https://doi.org/10.1007/s11119-020-09750-2

Lee, C., Strong, R., & Dooley, K. (2021). Analyzing Precision Agriculture Adoption across the Globe: A Systematic Review of Scholarship from 1999–2020. *Sustainability*, *13*(18). https://doi.org/10.3390/su131810295

Liu, W., Shao, X., Wu, C., & Qiao, P. (2021). A systematic literature review on applications of information and communication technologies and blockchain technologies for precision agriculture development. *Journal of Cleaner Production*, 298. https://doi.org/10.1016/j.jclepro.2021.126763

Lowenberg-DeBoer, J., & Erickson, B. (2019). Setting the Record Straight on Precision Agriculture Adoption. *Agronomy Journal*, *111*, 1552–1569. https://doi.org/10.2134/agronj2018.12.0779

Mathanker, S. (2021). A Precision Agriculture Technology Course for Regions with Lower Technology Adoption Levels. *Applied Engineering in Agriculture*, *37*(5), 871–877. https://doi.org/10.13031/aea.14669

Michels, M., von Hobe, C., & Musshoff, O. (2020). A trans-theoretical model for the adoption of drones by large-scale German farmers. *Journal of Rural Studies*, *75*, 80–88. https://doi.org/10.1016/j.jrurstud.2020.01.005

Ogundari, K., & Bolarinwa, O. (2018). Impact of agricultural innovation adoption: a meta-analysis. *Australian Journal of Agricultural and Resource Economics*, *62*(2), 217–236. https://doi.org/10.1111/1467-8489.12247

Osrof, H., Tan, C., Angappa, G., Yeo, S., & Tan, K. (2023). Adoption of smart farming technologies in field operations: A systematic review and future research agenda. *Technology in Society*, *75*. https://doi.org/10.1016/j.techsoc.2023.102400

Page, M., McKenzie, J., Bossuyt, P., Boutron, I., Hoffmann, T., Mulrow, C., Shamseer, L., Tetzlaff, J., Akl, E., Brennan, S., Chou, R., Glanville, J., Grimshaw, J., Hróbjartsson, A., Lalu, M., Li, T., Loder, E., Mayo-Wilson, E., McDonald, S., et al. (2021). The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *Bmj*, *372*. https://doi.org/10.1136/bmj.n71

Panityakul, T., Bumrungsup, C., & Knapp, G. (2013). On Estimating Residual Heterogeneity in Random-Effects Meta-Regression: A Comparative Study. *Journal of Statistical Theory and Applications*, *12*(3). https://doi.org/10.2991/jsta.2013.12.3.4

Pathak, H., Brown, P., & Best, T. (2019). A systematic literature review of the factors affecting the precision agriculture adoption process. *Precision Agriculture*, *20*(6), 1292–1316. https://doi.org/10.1007/s11119-019-09653-x

Paustian, M., & Theuvsen, L. (2017). Adoption of precision agriculture technologies by German crop farmers. *Precision Agriculture*, *18*(5), 701–716. https://doi.org/10.1007/s11119-016-9482-5

Peters, J., Sutton, A., Jones, D., Abrams, K., & Rushton, L. (2008). Contour-enhanced meta-analysis funnel plots help distinguish publication bias from other causes of asymmetry. *Journal of Clinical Epidemiology*, *61*(10), 991–996. https://doi.org/10.1016/j.jclinepi.2007.11.010

Piña, R., Lange, K., Machado, V., & Bratcher, C. (2023). Big data technology adoption in beef production. *Smart Agricultural Technology*, *5*. https://doi.org/10.1016/j.atech.2023.100235

Pivoto, D., Barham, B., Waquil, P., Foguesatto, C., Corte, V., Zhang, D., & Talamini, E. (2019). Factors influencing the adoption of smart farming by Brazilian grain farmers. *International Food and Agribusiness Management Review*, *22*(4), 571–588. https://doi.org/10.22434/IFAMR2018.0086

Rahm, M., & Huffman, W. (1984). The Adoption of Reduced Tillage: The Role of Human Capital and Other Variables. *American Journal of Agricultural Economics*, *66*(4), 405–413. https://doi.org/10.2307/1240918

Riley, R., Higgins, J., & Deeks, J. (2011). Interpretation of random effects meta-analyses. *Bmj*, *342*(102), d549–d549. https://doi.org/10.1136/bmj.d549

Rogers, E. (2003). Diffusion of Innovations (5th ed). Free Press. ISBN 9780743222099.

Rothstein, H., Sutton, A., & Borenstein, M. (Eds.). (2005). *Publication Bias in Meta-Analysis*. Wiley. https://doi.org/10.1002/0470870168

Schimmelpfennig, D. (2016). *Farm Profits and Adoption of Precision Agriculture. ERR-217*. U.S. Department of Agriculture, Economic Research Service.

Shamseer, L., Moher, D., Clarke, M., Ghersi, D., Liberati, A., Petticrew, M., Shekelle, P., & Stewart, L. (2015). Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015: elaboration and explanation. *Bmj*, *349*(021), g7647–g7647. https://doi.org/10.1136/bmj.g7647

Sood, A., Bhardwaj, A., & Sharma, R. (2022). Towards sustainable agriculture: key determinants of adopting artificial intelligence in agriculture. *Journal of Decision Systems*, *2022*, 1–45. https://doi.org/10.1080/12460125.2022.2154419

Tamirat, T., Pedersen, S., & Lind, K. (2017). Farm and operator characteristics affecting adoption of precision agriculture in Denmark and Germany. *Acta Agriculturae Scandinavica Section B — Soil & Plant Science*, *68*(4), 349–357. https://doi.org/10.1080/09064710.2017.1402949

Tey, Y., & Brindal, M. (2022). A meta-analysis of factors driving the adoption of precision agriculture. *Precision Agriculture*, *23*(2), 353–372. https://doi.org/10.1007/s11119-021-09840-9

Torero, M. (2021). Robotics and AI in Food Security and Innovation: Why They Matter and How to Harness Their Power. In J. von Braun, M. S. Archer, G. Reichberg & M. Sánchez Sorondo (Eds.), J. von Braun, M. S. Archer, G. Reichberg, M. Sánchez Sorondo, Robotics, AI, and Humanity (pp. 99–107). Springer International Publishing. https://doi.org/10.1007/978-3-030-54173-6_8

Troiano, S., Carzedda, M., & Marangon, F. (2023). Better richer than environmentally friendly? Describing preferences toward and factors affecting precision agriculture adoption in Italy. *Agricultural and Food Economics*, *11*(1). https://doi.org/10.1186/s40100-023-00247-w

Vecchio, Y., Agnusdei, G., Miglietta, P., & Capitanio, F. (2020). Adoption of Precision Farming Tools: The Case of Italian Farmers. *International Journal of Environmental Research and Public Health*, *17*(3). https://doi.org/10.3390/ijerph17030869

Venkatesh, V., Morris, M., Davis, G., & Davis, F. (2003). User Acceptance of Information Technology: Toward a Unified View. *MIS Quarterly*, *27*(3), 425–478. https://doi.org/10.2307/30036540

Veroniki, A., Jackson, D., Viechtbauer, W., Bender, R., Bowden, J., Knapp, G., Kuss, O., Higgins, J., Langan, D., & Salanti, G. (2016). Methods to estimate the between-study variance and its uncertainty in meta-analysis. *Research Synthesis Methods*, *7*(1), 55–79. https://doi.org/10.1002/jrsm.1164

Walton, J., Roberts, R., Lambert, D., Larson, J., English, B., Larkin, S., Martin, S., Marra, M., Paxton, K., & Reeves, J. (2010). Grid soil sampling adoption and abandonment in cotton production. *Precision Agriculture*, *11*(2), 135–147. https://doi.org/10.1007/s11119-009-9144-y

Yarashynskaya, A., & Prus, P. (2022). Precision Agriculture Implementation Factors and Adoption Potential: The Case Study of Polish Agriculture. *Agronomy*, *12*(9). https://doi.org/10.3390/agronomy12092226

Yatribi, T. (2020). Factors Affecting Precision Agriculture Adoption: A Systematic Litterature Review. *ECONOMICS*, *8*(2), 103–121. https://doi.org/10.2478/eoik-2020-0013

Yue, M., Li, W., Jin, S., Chen, J., Chang, Q., Glyn, J., Cao, Y., Yang, G., Li, Z., & Frewer, L. (2023). Farmers' precision pesticide technology adoption and its influencing factors: Evidence from apple production areas in China. *Journal of Integrative Agriculture*, *22*(1), 292–305. https://doi.org/10.1016/j.jia.2022.11.002

Zuo, A., Wheeler, S., & Sun, H. (2021). Flying over the farm: understanding drone adoption by Australian irrigators. *Precision Agriculture*, *22*(6), 1973–1991. https://doi.org/10.1007/s11119-021-09821-y

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.